

Signal Processing

Exercise Sheet 4 – Convolution

Summer Term 2026

Cheatsheet

- Convolution, convolution integral or convolution product

$$\begin{aligned}y(t) &= \int_{-\infty}^{+\infty} u(\tau) \cdot g(t - \tau) d\tau \\ &= u(t) * g(t)\end{aligned}$$

- General procedure for solving the convolution graphically
 1. Mirror one of the signals and “shift it” in your mind
 2. Identify interesting sections / boundaries
 3. Calculate the integral of the product
- General procedure for the mathematical calculation of the convolution
 1. Sketch both signals
 2. Decide which of the two signals can be mirrored and shifted “more easily”
 3. Define time intervals for case distinctions
 4. Draw a separate sketch for each case
 5. Define integration limits for each case
 6. Calculate the integration for each case
- What is the convolution used for? Let $h(t)$ be the impulse response of an energy-free LTI system.

$$y(t) = \int_{-\infty}^{+\infty} x(\tau) \cdot h(t - \tau) d\tau$$

→ The response $y(t)$ of an energy-free LTI system to any continuous input signal $x(t)$ can be determined unambiguously via the impulse response $h(t)$. (Proof for anyone interested on the following page.)

→ An unknown LTI system can be characterized by measuring the impulse response $h(t)$.

- Calculation rules for the convolution

Commutativity	$u(t) * g(t) = g(t) * u(t)$
Associativity	$x(t) * (u(t) * g(t)) = (x(t) * u(t)) * g(t)$
Distributivity	$(u_1(t) + u_2(t)) * g(t) = u_1(t) * g(t) + u_2(t) * g(t)$
Neutral element	$u(t) * \delta(t) = u(t)$ $u(t) * \delta(t - t_0) = u(t - t_0)$

Proof:

- The response of an LTI system $\mathfrak{S}$ to a Dirac impulse $\delta(t)$ is referred to as the impulse response $h(t)$.

$$h(t) = \mathfrak{S}(\delta(t))$$

- An arbitrary, continuous input signal $x(t)$ can be transformed as follows using the neutral element property of $\delta(t)$.

$$x(t) = \int_{-\infty}^{+\infty} x(\tau) \cdot \delta(t - \tau) d\tau$$

- The response of the system $\mathfrak{S}$ is thus:

$$y(t) = \mathfrak{S}(x(t)) = \mathfrak{S}\left(\int_{-\infty}^{+\infty} x(\tau) \cdot \delta(t - \tau) d\tau\right)$$

- From the linearity of the system follows (since $x(\tau)$ is not dependent on t):

$$y(t) = \mathfrak{S}(x(t)) = \int_{-\infty}^{+\infty} x(\tau) \cdot \mathfrak{S}(\delta(t - \tau)) d\tau$$

- The time invariance of the system finally leads (with $h(t) = \mathfrak{S}(\delta(t))$) to:

$$y(t) = \int_{-\infty}^{+\infty} x(\tau) \cdot h(t - \tau) d\tau$$

Question 1

For a system $\mathfrak{S}$, the impulse response $h(t)$ and the input signal $x(t)$ are given as:

$$h(t) = \text{rect}(t - 0.5)$$

$$x(t) = \sigma(t)$$

- Sketch the impulse response $h(t)$ and the input signal $x(t)$.
- Calculate the output signal $y(t)$ by convolution.
- Sketch the output signal $y(t)$.

Question 2

For a system $\mathfrak{S}$, the impulse response $h(t)$ and the input signal $x(t)$ are given as:

$$h(t) = \alpha \cdot \text{rect}\left(\frac{t - 1.5}{3}\right), \alpha \in \mathbb{R}^+$$

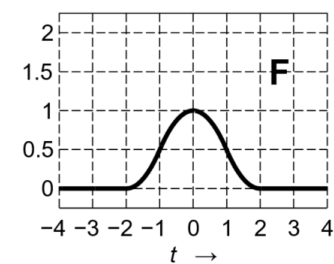
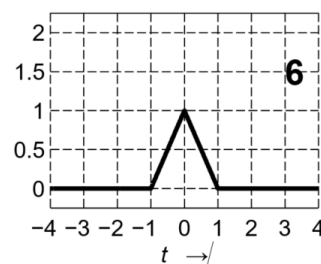
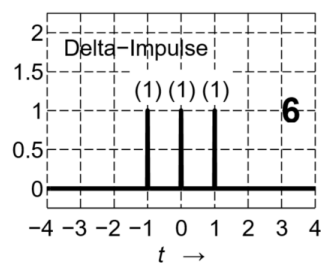
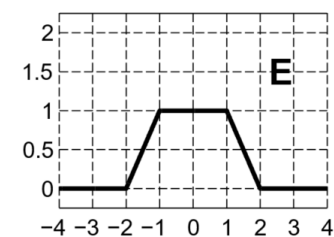
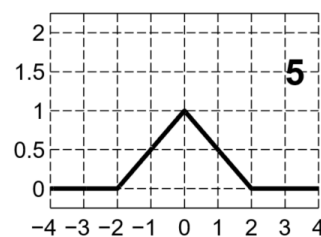
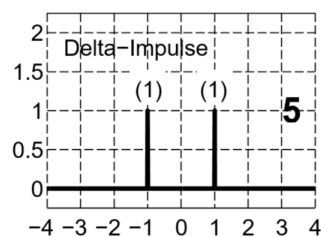
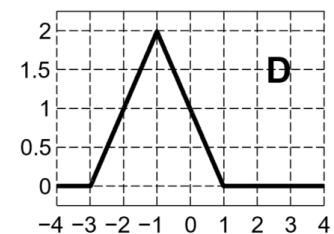
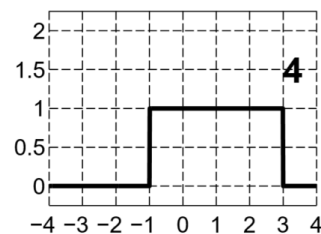
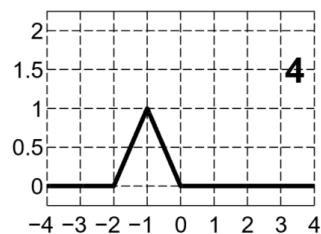
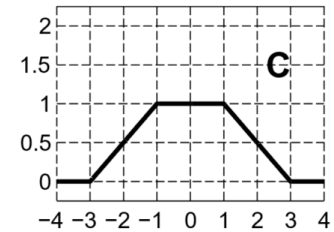
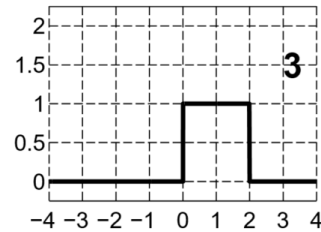
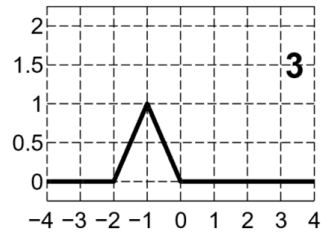
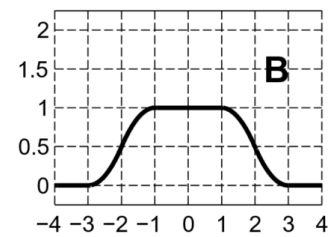
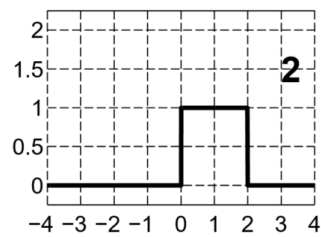
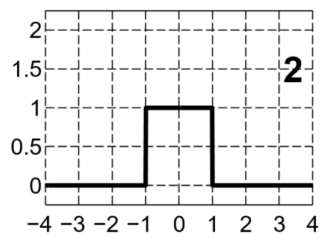
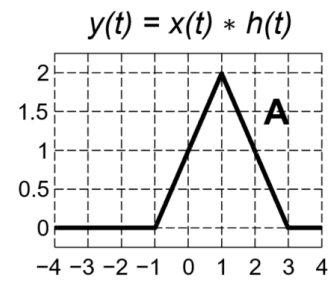
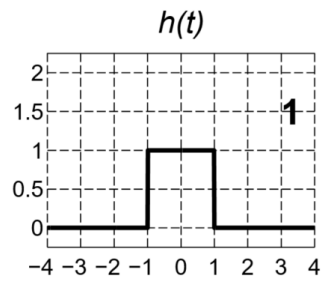
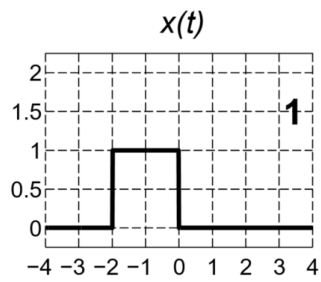
$$x(t) = t \cdot \text{rect}\left(\frac{t - 1}{2}\right)$$

Determine the output signal $y(t)$ by calculating the convolution of the two signals $x(t)$ and $h(t)$.

Question 3

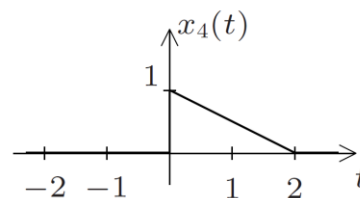
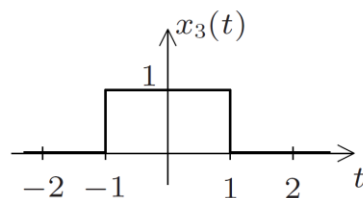
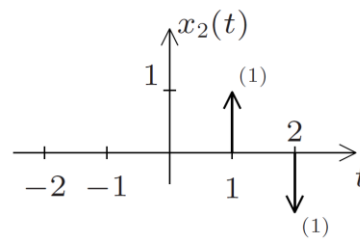
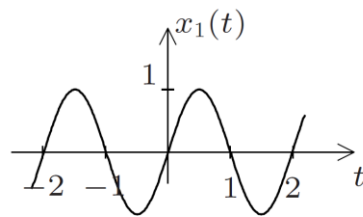
On the next page six pairs of signals $x(t)$ und $h(t)$ (1 - 6) are sketched. For each pair the convolution $y(t) = x(t) * h(t)$ (A - F) was calculated and sketched as well. They are not in the correct order.

Assign each pair $(x(t), h(t))$ to the corresponding convolution $(y(t))$ by giving a system theoretic justification.



Question 4

The signals $x_1(t)$ to $x_4(t)$ are given by:



Sketch the following convolutions. Calculations are not necessary.

- a) $x_1(t) * x_2(t)$
- b) $x_1(t) * x_3(t)$
- c) $x_2(t) * x_3(t)$
- d) $x_3(t) * x_3(t)$
- e) $x_2(t) * x_4(t)$
- f) $x_3(t) * x_4(t)$